Polycystic ovary syndrome

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Key facts

- **Polycystic ovary syndrome (PCOS) is a common hormonal disorder in which higher than normal androgen levels lead to irregular menstrual periods, abnormal ovulation, infertility, excess facial or body hair and/or acne in women (1).**
- **PCOS affects an estimated 10–13% of reproductive-aged women. It is estimated that up 70% of women with PCOS worldwide do not know they have this condition.**
- **PCOS is the most common cause of anovulation among women globally and a leading cause of infertility.**
- **PCOS runs in families but can cause a range of different symptoms and affect women in different ways. PCOS is a chronic metabolic condition that persists beyond the reproductive years.**
- **Women with PCOS are at higher risk for a variety of longer-term health problems that affect physical and emotional well-being compared to women without PCOS, including insulin resistance, type 2 diabetes mellitus and obesity.**

Overview

Polycystic ovary syndrome (PCOS) is a common hormonal disorder that affects women during their reproductive years and beyond, into later stages of life.

PCOS occurs when inappropriate hormonal signaling leads to higher than normal androgen levels and other hormonal imbalances. As a result, women with PCOS may experience irregular or infrequent menstrual periods, pain – including pain with heavy menstrual bleeding, abnormal ovulation, changes in hair (either excessive facial or body hair or female-pattern baldness), oilier skin, acne and/or cysts in the ovaries.

PCOS is a leading cause of irregular menstrual periods and one of the most common causes of infertility. Irregular or infrequent menstrual periods among women with PCOS may also raise their risk for endometrial hyperplasia and/or endometrial cancer. PCOS can start as early as during adolescence but is most commonly detected or diagnosed when women struggle to become pregnant.

Further, PCOS is a chronic metabolic condition associated with heightened long-term risk for insulin resistance, type 2 diabetes mellitus and obesity. There is currently no cure for PCOS, but lifestyle changes, medications and fertility treatments can reduce symptoms, increase fertility rates and protect longer-term health.

The cause of PCOS is unknown, but women with a family history of PCOS or type 2 diabetes are at higher risk.

Scope of the problem

PCOS a significant global public health problem and is one of the most common hormonal disorders affecting women, often becoming apparent during the reproductive years. An estimated 10–13% of women globally are thought to have PCOS, but up to 70% of affected women are undiagnosed.

The prevalence of PCOS may be higher among women in some racial or ethnic groups than others, and some women have higher genetic propensity for metabolic complications. However, structural racism and other social determinants of health shape the extent to which different groups of women with PCOS seek and receive quality health services.

Symptoms

Symptoms of PCOS can differ from person to person, and an individual's symptoms may change over time.

Possible symptoms include:

- **intermittent, unpredictable or absent periods; or, for some women, heavy or long and/or painful periods**
- **difficulty conceiving or infertility**
- **excessive hair on the face or body**
- **female-pattern baldness or hair thinning**
- **acne or oily skin.**

Women with PCOS have an increased risk of developing other health conditions including:

- **gestational diabetes or high blood pressure in pregnancy**
- **weight gain, especially around the belly**
- **type 2 diabetes**
- **hypertension (high blood pressure)**
- **high cholesterol**
- **cardiovascular disease**
- **obesity**
- **sleep apnea**
- **metabolic steatohepatitis**
- **endometrial hyperplasia or endometrial cancer (abnormal growth or cancer, respectively, of the inner lining of the uterus).**

PCOS can adversely impact quality of life in substantial ways, and women with PCOS may experience anxiety, depression, eating disorders and/or a negative body image. Symptoms such as infertility, obesity and unwanted hair growth are subject to social stigma in many settings. This can affect social and family relationships, work, a sense of belonging in one's community, mental health and other aspects of well-being.

Diagnosis

PCOS is diagnosed by the presence of at least two of the following, once other causes for these symptoms have been excluded:

1. **signs or symptoms of high androgens (excess facial or bodily hair, loss of hair from the head, acne or oily skin) or an elevated blood level of testosterone;**
2. **irregular or absent menstrual periods; and**
3. **polycystic ovaries on an ultrasound in women.**

This means that some women with PCOS do not have polycystic ovaries, and ovarian cysts are not required for PCOS diagnosis. Blood tests can be used to identify higher than normal levels of androgen steroids in women who do not have visible excess hair or acne or to check whether ovulation is occurring normally in women who might have PCOS but who have regular menstrual cycles. When making a PCOS diagnosis, health workers consider genetic risk factors since women with a family history of PCOS or type 2 diabetes are at higher risk of PCOS. Health workers also take life stage into account because irregular periods can be normal at either the beginning (menarche) or end (perimenopause) of the reproductive years.

Women who are newly diagnosed with PCOS are recommended to have additional blood testing to evaluate for underlying insulin resistance and to assess their overall cardiovascular risk.

Treatment

There is no cure for PCOS, but treatments can improve quality of life, assist with fertility, reduce the risk of endometrial hyperplasia and/or endometrial cancer and help prevent long-term cardiovascular events. Seeking health care early to manage symptoms of PCOS can help women protect their health and well-being during and beyond their reproductive years.

Healthy lifestyle behaviors, including healthy eating and physical activity, are important for all women with PCOS, even if they do not lead to weight loss.

Combined oral contraceptive pills can help regulate the menstrual cycle and may reduce acne or excess facial and body hair. Androgen receptor blockers may also be used to treat excess hair or acne. Treatment choices should be based on shared decision-making, taking every individual's unique values and preferences into account.

Treatments for infertility due to PCOS include lifestyle changes, medicines or surgery to stimulate ovulation. In-vitro fertilization (IVF) and other assistive reproductive technologies may also be used. Women with PCOS can be supported to become pregnant but generally have higher risk of pregnancy complications that warrants additional monitoring.

WHO response

WHO promotes research regarding prevention, diagnosis and treatment of infertility for women with PCOS as well as broader menstrual health, seeking to identify and elevate the most important unanswered related questions. WHO works with Member States and partners to address PCOS as part of its broad efforts to improve women's health and well-being globally. WHO works with governmental and nongovernmental partners to enhance public awareness related to PCOS, and to generate guidance for health-care providers on how it can be recognized and treated.

WHO supports the generation of statistics on how common infertility and its causes (such as PCOS) are. WHO also helps Member States to pay more attention to infertility and its causes within national health policies, services and financing to ensure that everyone who needs health care related to infertility is able to obtain it without financial hardship.

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1) In this fact sheet, the term "women" is used inclusively to refer to all women, girls and gender-diverse individuals who menstruate and who may therefore be at risk of developing PCOS at some point in their life.